

Module

2

Networking Basics

Theory



Training structure



Content

- Understanding how networks work
- Core of cybersecurity knowledge
- 32 hours: 10 theory, 22 practical
- Hands-on with real tools

Networking Theory



Why Networks Matter in

- Attacks travel through networks
- Monitor traffic = detect threats
- Understand protocols = find anomalies
- Network skills = SOC analyst success



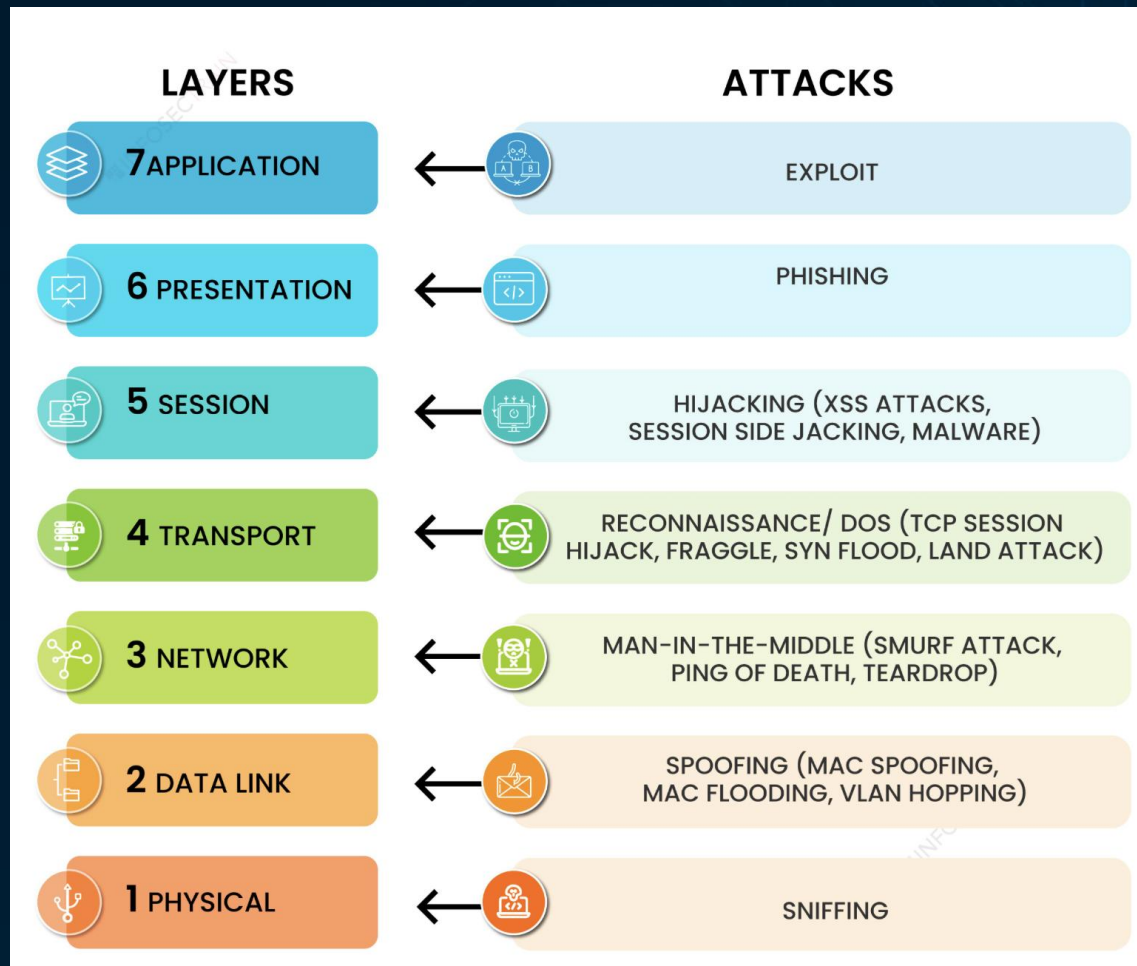
The OSI Model

Overview

- 7 layers of network communication:
- Application, Presentation, Session
- Transport, Network, Data Link, Physical

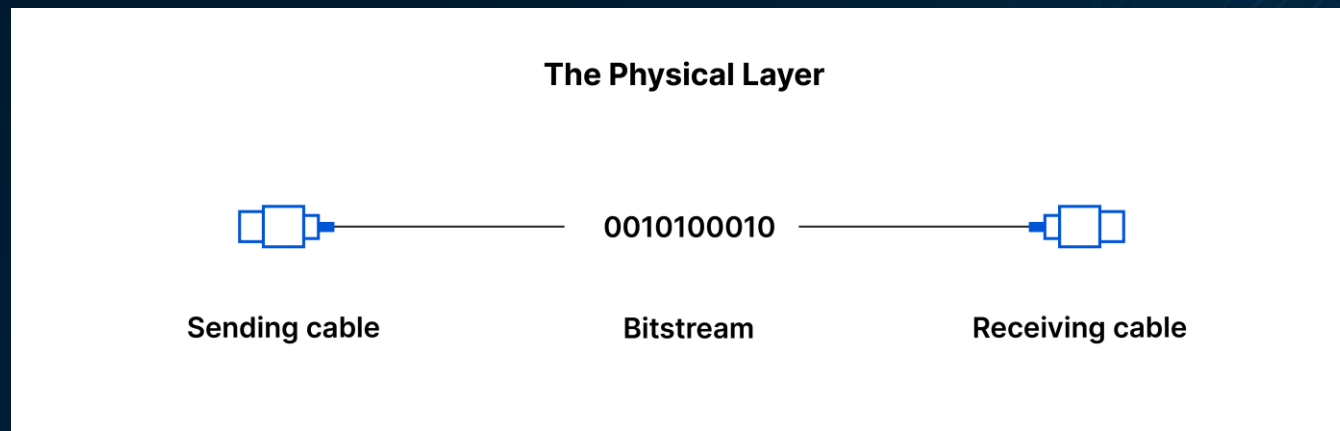
Memory trick: "All People Seem To Need Data Processing"

The OSI Model Overview With Attacks



OSI Layer 1 Physical

- Layer 1: Physical
- Actual hardware and signals
- Cables, fiber, radio waves
- Bits (1s and 0s)
- Examples: Ethernet cable, WiFi antenna

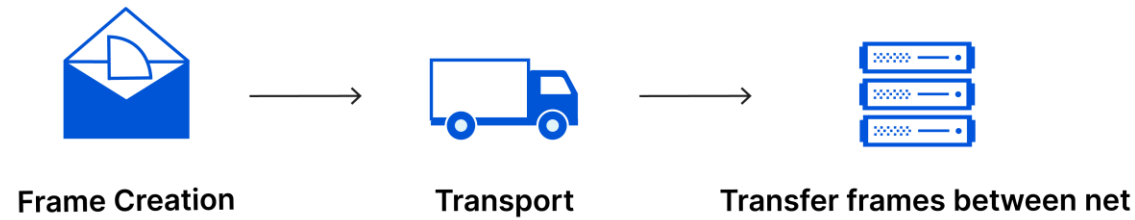




OSI Layer 2 Data Link

- Layer 2: Data Link
- MAC addresses (hardware addresses)
- Switches operate here
- Frame creation
- Examples: Ethernet, WiFi (802.11)

The Data Link Layer



OSI Layer 3 Network

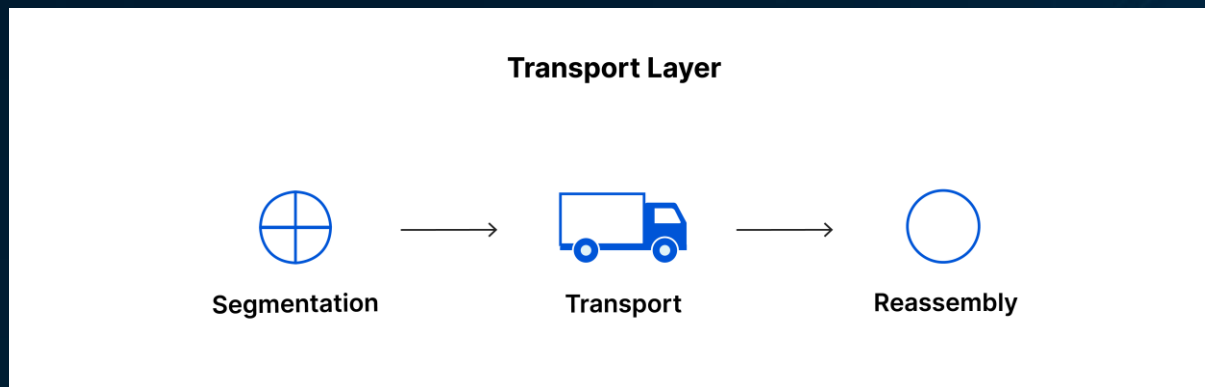
- Layer 3: Network
- IP addresses (logical addresses)
- Routers operate here
- Packet routing
- Examples: IPv4, IPv6, ICMP

The Network Layer



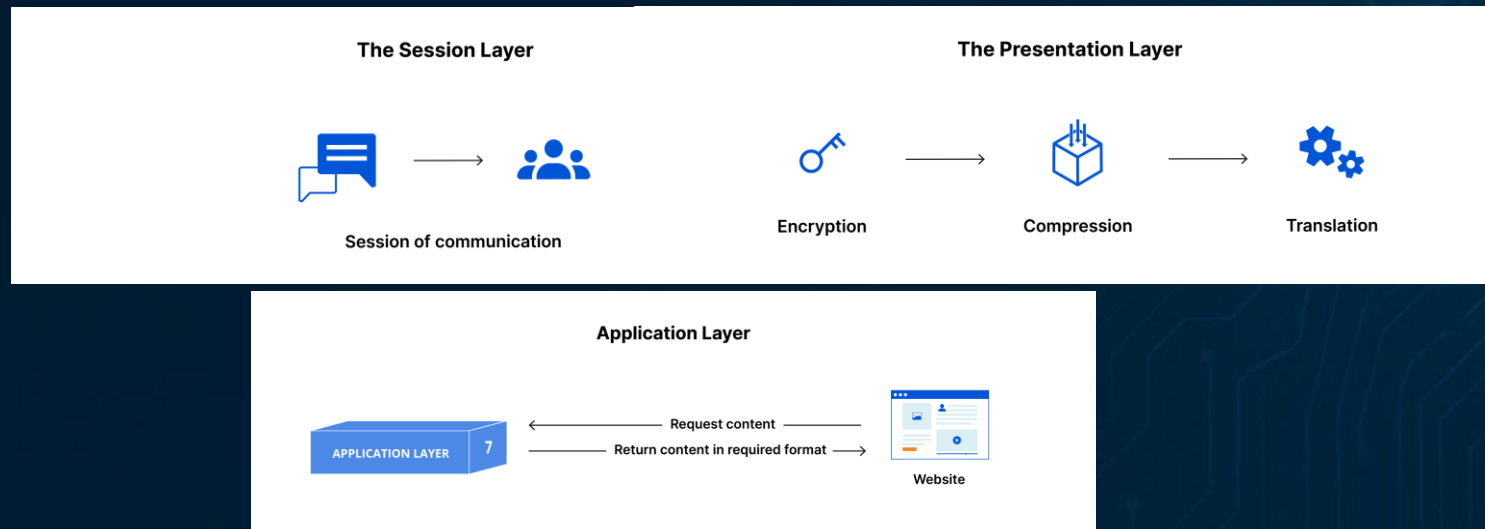
OSI Layer 4 Transport

- Layer 4: Transport
- Port numbers (application identifiers)
- TCP: Reliable, slower
- UDP: Fast, unreliable
- Examples: TCP port 80, UDP port 53



OSI Layers 5-7 Upper Layers

- Layer 5: Session (connections)
- Layer 6: Presentation (encryption, formatting)
- Layer 7: Application (user programs)
- Examples: HTTP, SMTP, DNS, SSH



OSI Model Summary

Layer	Name	Key Concept
7	Application	HTTP, DNS
6	Presentation	Encryption
5	Session	Connections
4	Transport	TCP/UDP ports
3	Network	IP addresses
2	Data Link	MAC addresses
1	Physical	Cables, signals

4-layer TCP/IP model compared to OSI

- TCP/IP Simplified model:
 - Application (OSI 5-7)
 - Transport (OSI 4)
 - Internet (OSI 3)
 - Network Access (OSI 1-2)

Encapsulation Process

- Data traveling down stack:
 - Layer 7: Data
 - Layer 4: Segment (+ port)
 - Layer 3: Packet (+ IP)
 - Layer 2: Frame (+ MAC)
 - Layer 1: Bits

Why Models Matter for Security

- Security at each layer:
 - Layer 7: Application attacks
 - Layer 4: Port scans
 - Layer 3: IP spoofing
 - Layer 2: ARP poisoning
 - Layer 1: Cable tapping

First Subject



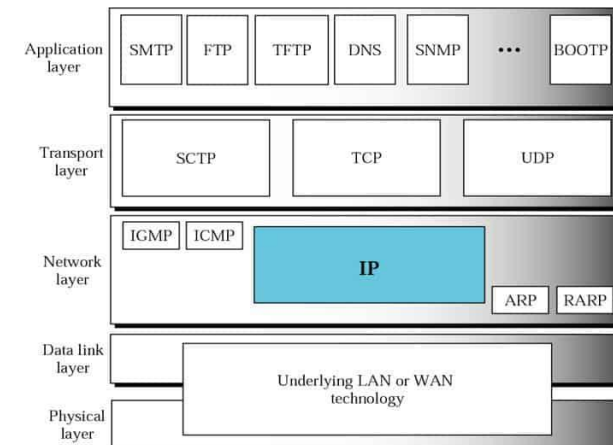
What is a Protocol?

- Agreed-upon communication rules
- Defines message format
- Specifies timing and order
- Like a language for computers

IP Internet Protocol

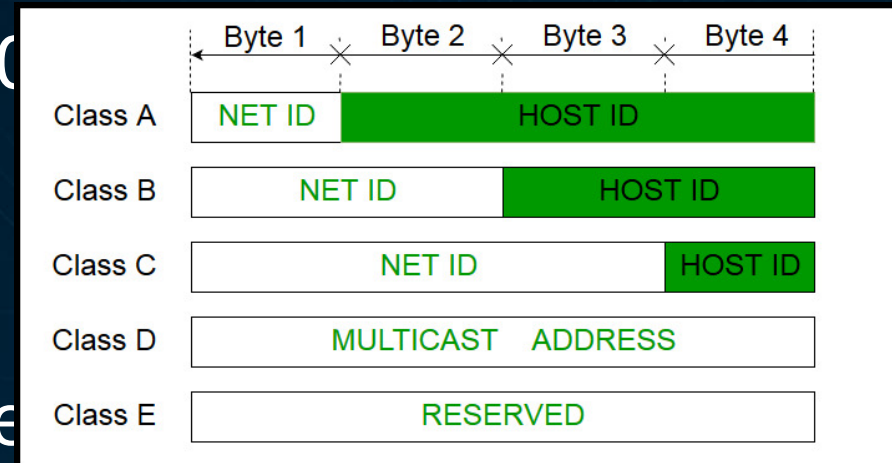
- IP (Internet Protocol)
- Layer 3 protocol
- Routes packets between networks
- Two versions: IPv4 and IPv6
- Example: 192.168.1.10

Position of IP in TCP/IP protocol suite



IPv4 Addressing

- IPv4 format: 192.168.1.10
- Four octets (0-255)
- 32 bits total
- About 4.3 billion addresses



- Classes: A, B, C (for different network sizes)

Special IP Addresses

- Private ranges (not routable):
 - 10.0.0.0 10.255.255.255
 - 172.16.0.0 172.31.255.255
 - 192.168.0.0 192.168.255.255
- Special:
 - 127.0.0.1 (localhost)
 - 0.0.0.0 (any address)



CYBERLABS

TCP Transmission Control Protocol

- TCP Reliable delivery:
 - Connection-oriented (handshake)
 - Guarantees delivery
 - Error checking
- Used for: Web, email, file transfer, SSH

```
03:06:31.546515 IP siftworkstation.ssh > 192.168.184.1.53548: Flags [P.], seq 41132:41504, ack 1801, win 761, options [nop,nop,TS val 18284
36746 ecr 2390120595], length 372
03:06:31.546991 IP 192.168.184.1.53548 > siftworkstation.ssh: Flags [.], ack 41504, win 2048, options [nop,nop,TS val 2390120698 ecr 182843
6746], length 0
03:06:31.651585 IP siftworkstation.ssh > 192.168.184.1.53548: Flags [P.], seq 41504:41876, ack 1801, win 761, options [nop,nop,TS val 18284
36851 ecr 2390120698], length 372
```

UDP User Datagram Protocol

- UDP Fast but unreliable:
 - Connectionless (no handshake)
 - No delivery guarantee
 - Faster than TCP
- Used for: Streaming, gaming, DNS, Service Discovery

```
03:11:38.796126 IP 192.168.184.1.57621 > 192.168.184.255.57621: UDP, length 44
03:11:54.273961 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:55.274901 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:56.276083 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:57.276958 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:12:08.135741 IP 192.168.184.1.60134 > 239.255.255.250.1900: UDP, length 125
```

Ports Subject



Port Numbers

- Ports identify applications:
 - 0-1023: Well-known ports
 - 1024-49151: Registered ports
 - 49152-65535: Dynamic/ephemeral

Think: Apartment building IP is street address,
port is apartment number. On top are hotel rooms.

Common Ports You Must Know

Port	Protocol	Service
20/21	FTP	File transfer service
22	SSH	Secure Shell
25	SMTP	Email sending (unencrypted!)
53	DNS	Domain Names
80	HTTP	Web (unencrypted!)
443	HTTPS	Web (encrypted)
3389	RDP	Windows Remote Desktop

ICMP Internet Control Message Protocol

- ICMP Network diagnostics:

- Layer 3 protocol
- No ports (not TCP/UDP)
- Error reporting

- Tools: ping, traceroute

```
ping nu.nl
```

```
PING nu.nl (118.69.16.182) 56(84) bytes of data.  
64 bytes from 118.69.16.182: icmp_seq=1 ttl=128 time=6.31 ms  
64 bytes from 118.69.16.182: icmp_seq=2 ttl=128 time=6.21 ms  
64 bytes from 118.69.16.182: icmp_seq=3 ttl=128 time=4.78 ms
```

```
03:06:30.091568 IP siftworkstation > 118.69.16.182: ICMP echo request, id 21023, seq 7, length 64  
03:06:30.096473 IP 118.69.16.182 > siftworkstation: ICMP echo reply, id 21023, seq 7, length 64
```

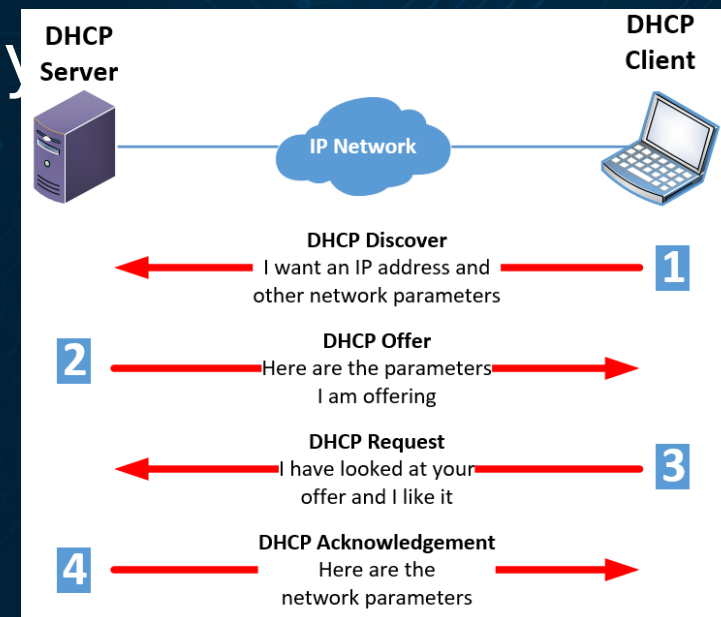
ARP Address Resolution Protocol

- ARP Maps IP to MAC:
 - Operates at Layer 2
 - "Who has this IP address?"
 - Builds ARP cache/table
- Security risk: ARP spoofing

```
03:05:32.714613 ARP, Request who-has _gateway tell siftworkstation, length 28
03:05:32.714860 ARP, Request who-has 192.168.184.254 tell siftworkstation, length 28
03:05:32.714963 ARP, Reply _gateway is-at 00:50:56:f1:6b:6e (oui Unknown), length 46
03:05:32.715170 ARP, Reply 192.168.184.254 is-at 00:50:56:ef:b3:d3 (oui Unknown), length 46
```

DHCP Dynamic Host Configuration Protocol

- DHCP Automatic IP assignment:
 - DORA process (Discover, Offer, Request, Acknowledge)
- Assigns IP, subnet, gateway
 - Temporary leases
 - Ports: UDP 67/68



DNS Domain Name System

- DNS Internet's phonebook:
 - Converts names to IPs
 - Hierarchical system
- Port: UDP 53 (TCP for large queries)
 - Example: google.com → 142.250.80.46

HTTP vs HTTPS

- HTTP HyperText Transfer Protocol:
 - Port 80
 - Unencrypted ⚠️
 - Plaintext visible
- HTTPS HTTP Secure:
 - Port 443
 - Encrypted with TLS/SSL ✅
 - Protects privacy

SSH Secure Shell

- SSH Secure remote access:
 - Port 22
 - Encrypted command line
 - Replaces insecure Telnet
 - Used for: Linux administration, file transfer

SMTP, POP3, IMAP Email Protocols

- Email protocols:
 - SMTP (25): Sending email
 - POP3 (110): Download & delete
 - IMAP (143): Sync across devices
- Secure versions: 465/587 (SMTPS), 995 (POP3S), 993 (IMAPS)

SNMP Simple Network Management Protocol

- SNMP Network device management:
 - Monitor routers, switches, servers
 - Ports: UDP 161/162
 - Versions: v1, v2c, v3
 - Security: Use SNMPv3 (encrypted)

Protocol Summary Security Perspective

- Secure (encrypted):
-  HTTPS, SSH, SFTP, IMAPS
- Insecure (plaintext):
-  HTTP, Telnet, FTP, SMTP
- Reconnaissance:
-  ICMP, DNS, SNMP

Next Week



Network Tools